

$$f : \mathbb{R}^3 \rightarrow \mathbb{R}$$

$$\begin{aligned}\max f(x) &= 4x_1 + x_2 - x_3 \\ x_1 + x_2 + x_3 &\leq 12 \\ 2x_1 + x_3 &\geq 6 \\ x_2 + x_3 &= 5 \\ x_1, x_2, x_3 &\geq 0\end{aligned}$$

Metodo delle due fasi

Fase I

$$\min f(x) = a_1 + a_2$$

$$\begin{aligned}x_1 + x_2 + x_3 + s_1 &= 12 \\ 2x_1 + x_3 - s_2 + a_1 &= 6 \\ x_2 + x_3 + a_2 &= 5 \\ x_1, x_2, x_3, s_1, s_2, a_1, a_2 &\geq 0\end{aligned}$$

Calcolo dei costi

$$c_i = c_N - c_B^T \cdot B_N^{-1}$$

$$c_1 = 0 - (0; 1; 1) \cdot \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} = -2$$

$$c_2 = 0 - (0; 1; 1) \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = -1$$

$$c_3 = 0 - (0; 1; 1) \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = -2$$

$$c_4 = 0 - (0; 1; 1) \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = 0$$

$$c_5 = 0 - (0; 1; 1) \cdot \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix} = 1$$

$$c_6 = 1 - (0; 1; 1) \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = 0$$

$$c_7 = 1 - (0; 1; 1) \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = 0$$

$$c_8 = 0 - (0; 1; 1) \cdot \begin{pmatrix} 12 \\ 6 \\ 5 \end{pmatrix} = -11$$

Forma Standard

$$\begin{aligned}\min f(x) &= -4x_1 - x_2 + x_3 \\ x_1 + x_2 + x_3 + s_1 &= 12 \\ 2x_1 + x_3 - s_2 &= 6 \\ x_2 + x_3 &= 5 \\ x_1, x_2, x_3, s_1, s_2 &\geq 0\end{aligned}$$

Metodo del simplesso

	x_1	x_2	x_3	s_1	s_2	a_1	a_2	b
s_1	1	1	1	1	0	0	0	12
a_1	2	0	1	0	-1	1	0	6
a_2	0	1	1	0	0	0	1	5
	-2	-1	-2	0	1	0	0	-11

Entra x_1 ed esce a_1 :

	x_1	x_2	x_3	s_1	s_2	a_1	a_2	b
s_1	1	1	1	1	0	0	0	12
a_1	2	0	1	0	-1	1	0	6
a_2	0	1	1	0	0	0	1	5
	-2	-1	-2	0	1	0	0	-11

$R4 \leftarrow R4 + R2$

$$\begin{array}{ccccccc|c} -2 & -1 & -2 & 0 & 1 & 0 & 0 & -11 \\ 2 & 0 & 1 & 0 & -1 & 1 & 0 & 6 \\ \hline 0 & -1 & -1 & 0 & 0 & 1 & 0 & -5 \end{array}$$

↓

	x_1	x_2	x_3	s_1	s_2	a_1	a_2	b
s_1	1	1	1	1	0	0	0	12
a_1	2	0	1	0	-1	1	0	6
a_2	0	1	1	0	0	0	1	5
	0	-1	-1	0	0	1	0	-5

$R2 \leftarrow R2 \cdot \frac{1}{2}$

	x_1	x_2	x_3	s_1	s_2	a_1	a_2	b
s_1	1	1	1	1	0	0	0	12
a_1	1	0	$\frac{1}{2}$	0	$-\frac{1}{2}$	$\frac{1}{2}$	0	3
a_2	0	1	1	0	0	0	1	5
	0	-1	-1	0	0	1	0	-5

$R1 \leftarrow R1 - R2$

$$\begin{array}{ccccccc|c} 1 & 1 & 1 & 1 & 0 & 0 & 0 & 12 \\ 1 & 0 & \frac{1}{2} & 0 & -\frac{1}{2} & \frac{1}{2} & 0 & 3 \\ \hline 0 & 1 & \frac{1}{2} & 1 & \frac{1}{2} & \frac{1}{2} & 0 & 9 \end{array}$$

↓

	x_1	x_2	x_3	s_1	s_2	a_1	a_2	b
s_1	0	1	$\frac{1}{2}$	1	$\frac{1}{2}$	$\frac{1}{2}$	0	9
x_1	1	0	$\frac{1}{2}$	0	$-\frac{1}{2}$	$\frac{1}{2}$	0	3
a_2	0	1	1	0	0	0	1	5
	0	-1	-1	0	0	1	0	-5

Entra x_2 ed esce a_2 :

	x_1	x_2	x_3	s_1	s_2	a_1	a_2	b
s_1	0	1	$\frac{1}{2}$	1	$\frac{1}{2}$	$\frac{1}{2}$	0	9
x_1	1	0	$\frac{1}{2}$	0	$-\frac{1}{2}$	$\frac{1}{2}$	0	3
a_2	0	1	1	0	0	0	1	5
	0	-1	-1	0	0	1	0	-5

$R4 \leftarrow R4 + R3$

$$\begin{array}{cccccccc|c} 0 & -1 & -1 & 0 & 0 & 1 & 0 & -5 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 & 5 \\ \hline 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \end{array}$$

↓

	x_1	x_2	x_3	s_1	s_2	a_1	a_2	b
s_1	0	1	$\frac{1}{2}$	1	$\frac{1}{2}$	$\frac{1}{2}$	0	9
x_1	1	0	$\frac{1}{2}$	0	$-\frac{1}{2}$	$\frac{1}{2}$	0	3
a_2	0	1	1	0	0	0	1	5
	0	0	0	0	0	1	1	0

$R1 \leftarrow R1 - R3$

$$\begin{array}{cccccccc|c} 0 & 1 & \frac{1}{2} & 1 & \frac{1}{2} & \frac{1}{2} & 0 & 9 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 & 5 \\ \hline 0 & 0 & -\frac{1}{2} & 1 & \frac{1}{2} & \frac{1}{2} & -1 & 4 \end{array}$$

↓

	x_1	x_2	x_3	s_1	s_2	a_1	a_2	b
s_1	0	0	$-\frac{1}{2}$	1	$\frac{1}{2}$	$\frac{1}{2}$	-1	4
x_1	1	0	$\frac{1}{2}$	0	$-\frac{1}{2}$	$\frac{1}{2}$	0	3
x_2	0	1	1	0	0	0	1	5
	0	0	0	0	0	1	1	0

$z = 0$ si passa alla seconda fase.

Fase II

$$\min f(x) = -4x_1 - x_2 + x_3$$

$$-\frac{1}{2}x_3 + s_1 + \frac{1}{2}s_2 + \frac{1}{2}a_1 - a_2 = 4$$

$$x_1 + \frac{1}{2}x_3 - \frac{1}{2}s_2 + \frac{1}{2}a_1 = 3$$

$$x_2 + x_3 + a_2 = 5$$

$$x_1, x_2, x_3, s_1, s_2, a_1, a_2 \geq 0$$

Si eliminano le variabili artificiali introdotte nella Fase I

$$\min f(x) = -4x_1 - x_2 + x_3$$

$$-\frac{1}{2}x_3 + s_1 + \frac{1}{2}s_2 = 4$$

$$x_1 + \frac{1}{2}x_3 - \frac{1}{2}s_2 = 3$$

$$x_2 + x_3 = 5$$

$$x_1, x_2, x_3, s_1, s_2 \geq 0$$

Calcolo dei costi della nuova tabella:

$$c_1 = -4 - (0; -4; -1) \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = 0$$

$$c_2 = -1 - (0; -4; -1) \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = 0$$

$$c_3 = 0 - (0; -4; -1) \cdot \begin{pmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ 1 \end{pmatrix} = 3$$

$$c_4 = 0 - (0; -4; -1) \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = 0$$

$$c_5 = 0 - (0; -4; -1) \cdot \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 0 \end{pmatrix} = -2$$

$$c_6 = 0 - (0; -4; -1) \cdot \begin{pmatrix} 4 \\ 3 \\ 5 \end{pmatrix} = 17$$

Tabella risultante:

	x_1	x_2	x_3	s_1	s_2	b
s_1	0	0	$-\frac{1}{2}$	1	$\frac{1}{2}$	4
x_1	1	0	$\frac{1}{2}$	0	$-\frac{1}{2}$	3
x_2	0	1	1	0	0	5
	0	0	3	0	-2	17

Entra s_2 ed esce s_1 :

	x_1	x_2	x_3	s_1	s_2	b
s_1	0	0	$-\frac{1}{2}$	1	$\frac{1}{2}$	4
x_1	1	0	$\frac{1}{2}$	0	$-\frac{1}{2}$	3
x_2	0	1	1	0	0	5
	0	0	3	0	-2	17

$R2 \leftarrow R2 + R1$

$$\begin{array}{cccccc|c} 1 & 0 & \frac{1}{2} & 0 & -\frac{1}{2} & 3 \\ 0 & 0 & -\frac{1}{2} & 1 & \frac{1}{2} & 4 \\ \hline 1 & 0 & 0 & 1 & 0 & 7 \end{array}$$

$\downarrow$

	x_1	x_2	x_3	s_1	s_2	b
s_1	0	0	$-\frac{1}{2}$	1	$\frac{1}{2}$	4
x_1	1	0	0	1	0	7
x_2	0	1	1	0	0	5
	0	0	3	0	-2	17

$$R1 \leftarrow R1 \cdot 2$$

	x_1	x_2	x_3	s_1	s_2	b
s_1	0	0	-1	2	1	8
x_1	1	0	0	1	0	7
x_2	0	1	1	0	0	5
	0	0	3	0	-2	17

$$R4 \leftarrow R4 + R1 \cdot 2$$

$$\begin{array}{cccccc|c} 0 & 0 & 3 & 0 & -2 & & 17 \\ 0 & 0 & -2 & 4 & 2 & & 16 \\ \hline 0 & 0 & 1 & 4 & 0 & & 33 \end{array}$$

$\downarrow$

	x_1	x_2	x_3	s_1	s_2	b
s_2	0	0	-1	2	1	8
x_1	1	0	0	1	0	7
x_2	0	1	1	0	0	5
	0	0	1	4	0	33

$$x_1 = 7$$

$$x_2 = 5$$

$$x_3 = 0$$

$$s_1 = 0$$

$$s_2 = 8$$

$$\min f(x) = -4x_1 - x_2 + x_3 \rightarrow f(x) = -4(7) - (5) + (0) = -28 - 5 = -33$$

$$\max f(x) = 4x_1 + x_2 - x_3 \rightarrow \boxed{33}$$